MoveIt 2

> **Warning**
>
> This tutorial is currently not functioning but is anticipated to be operational again in the near future.

Learning Objectives

This tutorial is supplementary to the Tutorials provided on the MoveIt Humble website.

Getting Started

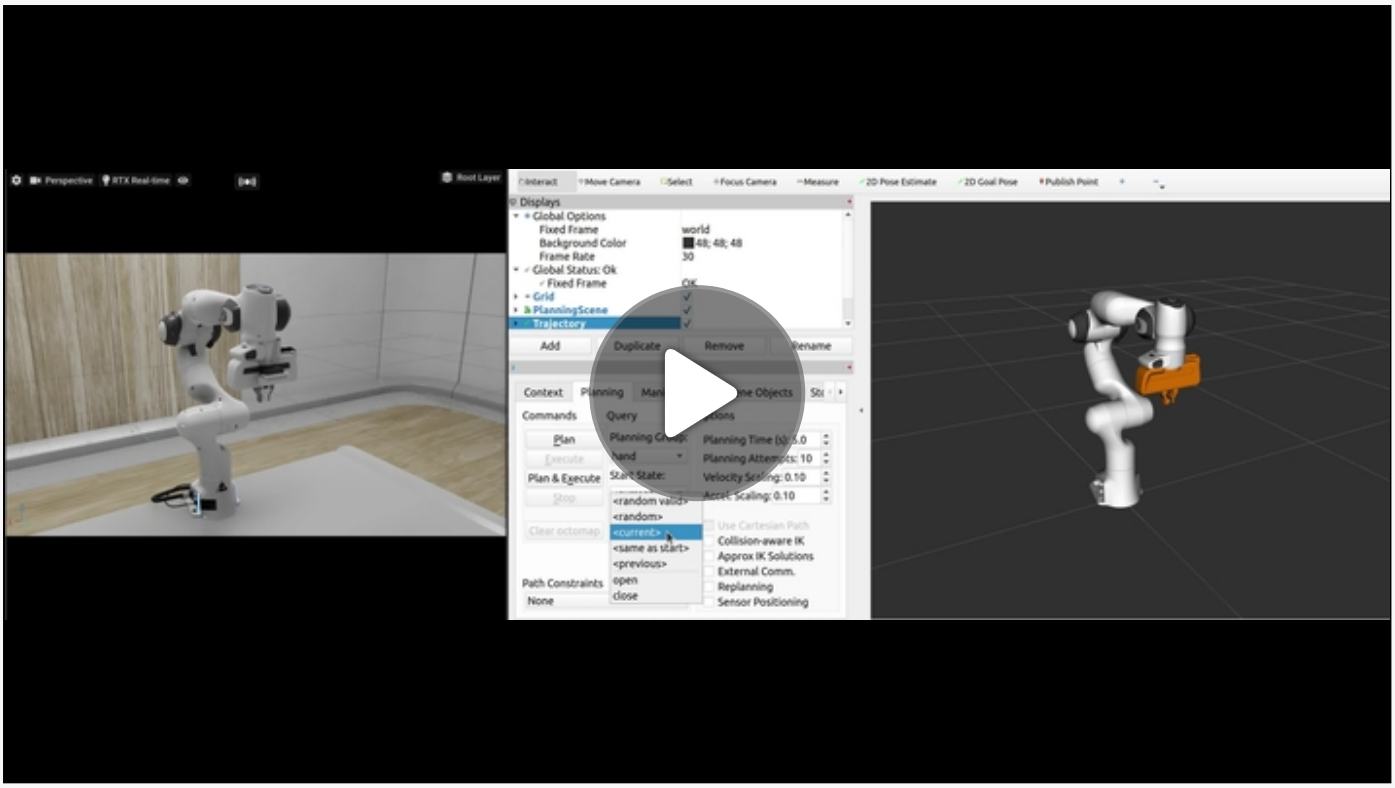
> **Important**
>
> Make sure to source your ROS 2 installation from the terminal before running Isaac Sim. If sourcing ROS 2 is a part of your `bashrc` then Isaac Sim can be run directly.

Prerequisite

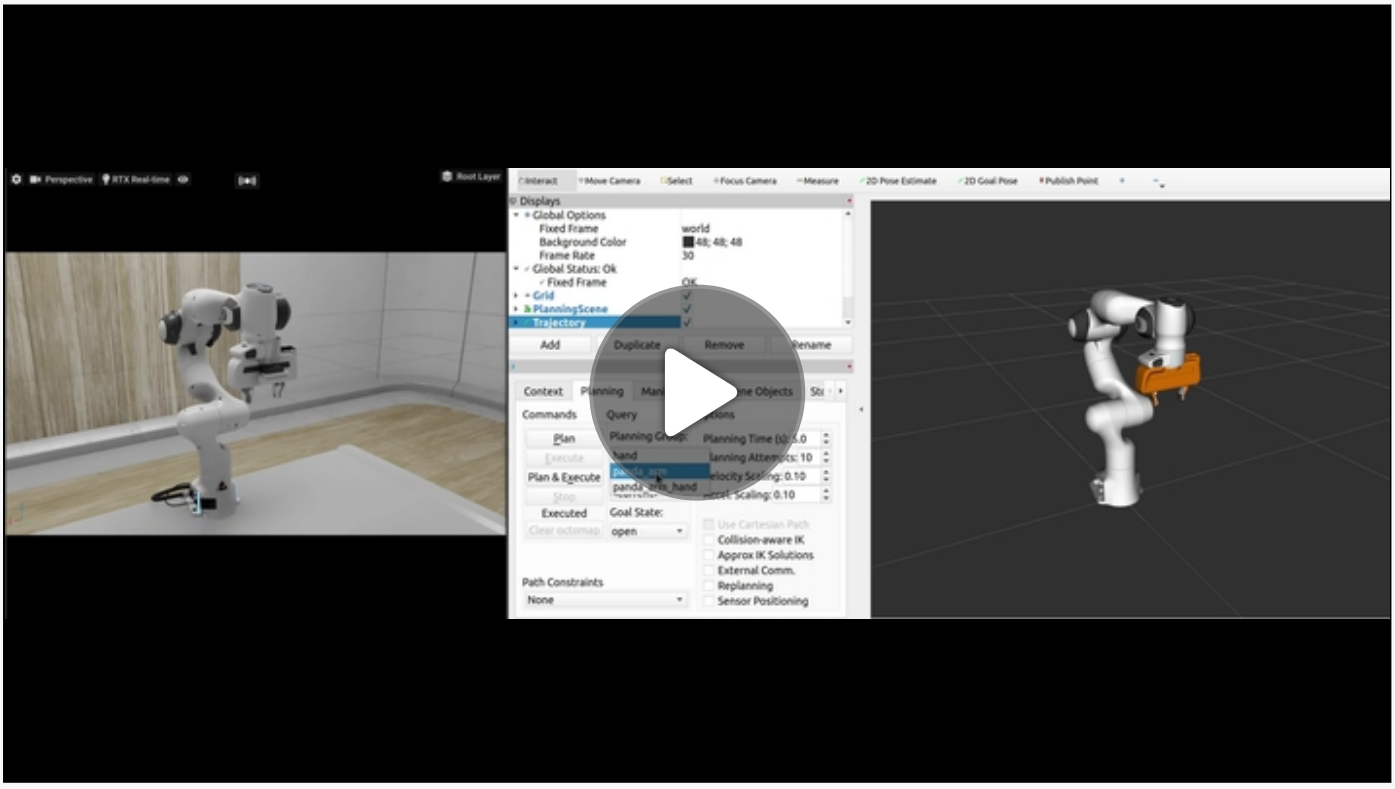
- Complete ROS and ROS 2 Installation.
- If using multiple systems, set the `FASTRTPS_DEFAULT_PROFILES_FILE` environment variable as per instructions in ROS and ROS 2 Installation before launching Isaac Sim, as well as any terminal where ROS messages will be sent or received, and ROS2 Extension is enabled.
- Completed ROS2 Joint Control: Extension Python Scripting.

Running MoveIt 2 Tutorial in Docker

1. Follow the instructions on the MoveIt Humble website to clone the repo, build the Docker, and launch both Isaac Sim and MoveIt via Python scripts.
2. Once Rviz is launched, you can start playing with the planner. Under `Planning Group`, the `hand` option should be selected. Under `Goal State`, select `open`.
3. Under **Commands**, click **Plan**. The planned movement of the hand will now be visualized.
4. Click **Execute**. The hand will start moving as planned earlier.



5. To plan the movement for the arm, under `Planning Group`, select `panda_arm`. Use the displayed arrows and rotation disks to set a goal position for the robot. Alternatively you can choose to select `<random_valid>` under `Goal State`.
6. Under **Commands**, click on **Plan** followed by **Execute** to visualize the planned motion of the arm and then move it.



Running MoveIt 2 from Examples Menu

You can also load the tutorial environment directly from example menu in Isaac Sim, as oppose to launching it from a standalone script.

1. Start an instance of Isaac Sim. Make sure the `FASTRTPS_DEFAULT_PROFILES_FILE` environment variable has been set.
2. Enable the `isaacsim.ros2.bridge` extension from the extension manager menu **Window > Extensions**.
3. Start with the same steps in MoveIt Tutorial. But instead of step 2, where the standalone Python script is launched, load the environment by going to **Window > Examples > Robotics Examples**, and then click on the **Robotics Examples** tab and expand the sections on the left hand side and open the example: **ROS2 > MoveIt > Franka MoveIt**.
4. Then continue on to step 3 and execute the Docker command to launch MoveIt2.

TroubleShooting

If your Rviz window is showing a black screen for where the robot should be, you may need to update your mesa driver. Add the following commands to `moveit2_tutorials/doc/how_to_guides/isaac_panda/.docker/Dockerfile` after line 17.

```
# update mesa driver
RUN apt update && apt install -y software-properties-common && add-apt-repository ppa:kis
RUN apt -y upgrade
```

Summary

Tips for running MoveIt2's Isaac Sim tutorial.

Next Steps

Continue on to the next tutorial in our ROS2 Tutorials series, ROS 2 Generic Publisher and Subscriber to learn how to publish and subscribe to and from any ROS 2 topic.

Further Learning

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